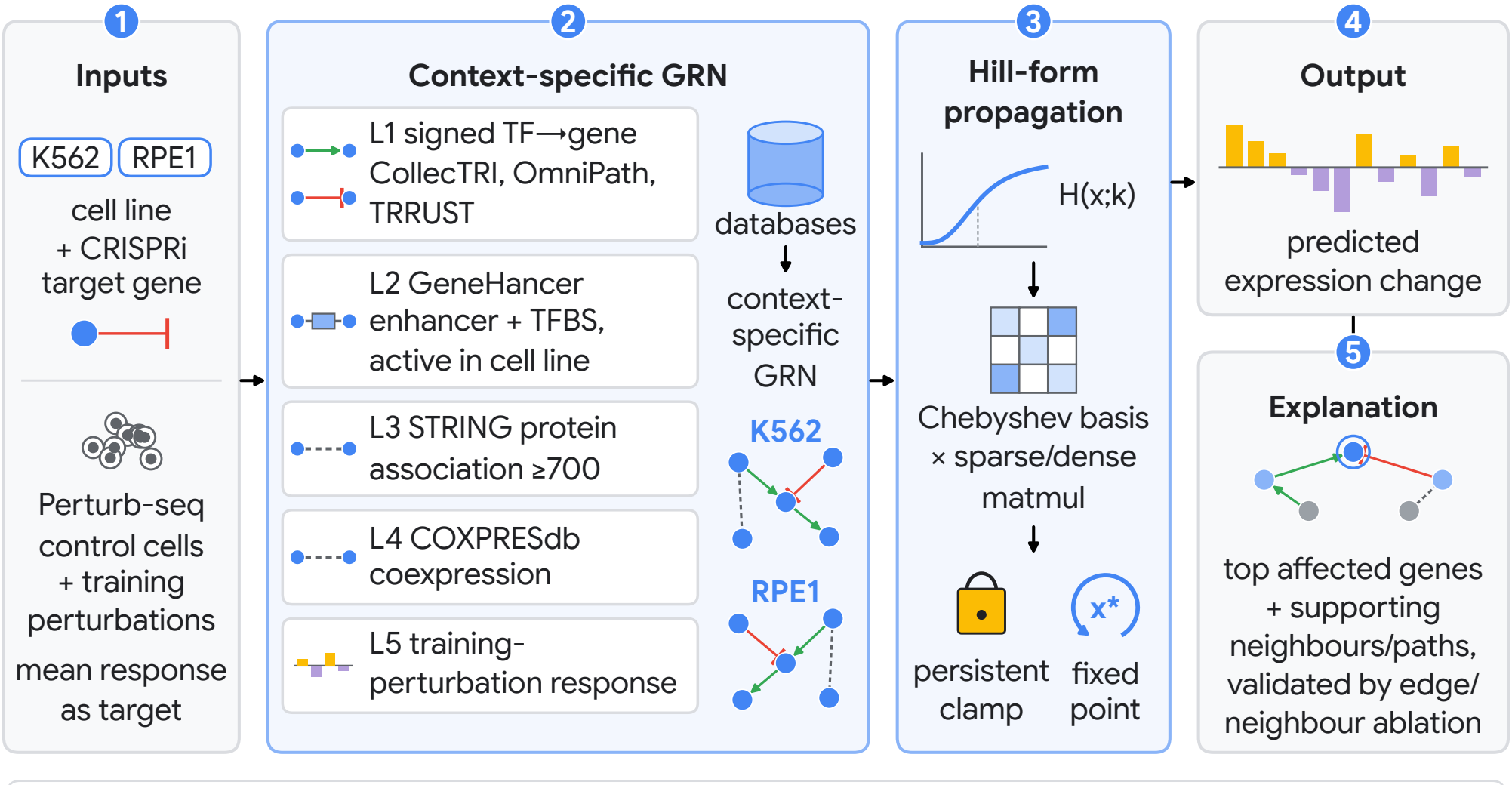
1
Inputs
K562
RPE1
cell line
+ CRISPRi
target gene
Perturb-seq
control cells
+ training
perturbations
mean response
as target
2
Context-specific GRN
L1 signed TF→gene
CollecTRI, OmniPath,
TRRUST
L2 GeneHancer
enhancer + TFBS,
active in cell line
L3 STRING protein
association ≥700
L4 COXPRESdb
coexpression
L5 training-
perturbation response
databases
context-
specific
GRN
K562
RPE1
3
Hill-form
propagation
H(x;k)
Chebyshev basis
× sparse/dense
matmul
persistent
clamp
x*
fixed
point
4
Output
predicted
expression change
5
Explanation
top affected genes
+ supporting
neighbours/paths,
validated by edge/
neighbour ablation
signed regulation
association (undirected)
data flow